Olist Commerce360

An End-to-End E-Commerce Data Warehouse and Business Intelligence Platform

P1_03 - Business Questions

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1. Introduction

This document lists 13 analytical questions (the course requires at least 10). Each question is linked to a business requirement in P1_02 and can be answered from the proposed warehouse. Each entry states the measures, dimensions, fact table, filters and data limitations. Measures are defined in P1_02 section 5.

2. Summary

ID	Question	Fact table(s)	Req.
BQ-01	What are monthly item sales revenue and order counts, and what is the month-over-month growth?	fact_order_items, fact_orders	BR-01
BQ-02	Which product categories generate the highest item sales revenue and units, and what share of the total does each represent?	fact_order_items	BR-02
BQ-03	Which customer regions and states generate the most revenue and orders, and which have the lowest average order value?	fact_orders	BR-03
BQ-04	How concentrated is revenue among sellers (share of the top 10% of sellers), and in which states are the top sellers located?	fact_order_items	BR-04
BQ-05	What is the payment method mix by number of payments and by amount paid, and how does it change over time?	fact_payments	BR-05
BQ-06	How does the average credit-card payment value change with the number of instalments chosen?	fact_payments	BR-05
BQ-07	What share of orders end in each status, and how does the cancellation rate change by month?	fact_orders	BR-06
BQ-08	What is the on-time delivery rate and the average delivery time by customer state and by month?	fact_orders	BR-07
BQ-09	Which fulfilment stage (approval, seller handling, carrier transit) contributes most to total delivery time, and does this differ by customer region?	fact_orders	BR-07
BQ-10	What is the distribution of review scores, and how does the average score and low-score share change by month?	fact_reviews	BR-08
BQ-11	Do orders delivered late receive lower average review scores than orders delivered on time?	fact_orders	BR-09
BQ-12	What share of customers made more than one purchase, and how much revenue comes from repeat orders?	fact_orders	BR-10
BQ-13	How large is freight relative to item price by product category, weight band and customer region?	fact_order_items	BR-11

3. Question details

BQ-01: What are monthly item sales revenue and order counts, and what is the month-over-month growth?

Business purpose	Track overall business trend and seasonality (BR-01).
Measures	Item sales revenue; Order count
Dimensions	dim_date (year > quarter > month)
Fact table(s)	fact_order_items, fact_orders
Filters / period	Purchase date; Jan 2017 - Aug 2018 for trends; exclude Cancelled status group.
Data limitation	Sep-Dec 2016 and Sep-Oct 2018 contain very few orders and are excluded from growth rates.
Requirement	BR-01

BQ-02: Which product categories generate the highest item sales revenue and units, and what share of the total does each represent?

Business purpose	Prioritise categories for assortment and marketing (BR-02).
Measures	Item sales revenue; Units sold
Dimensions	dim_product (category > product); dim_date
Fact table(s)	fact_order_items
Filters / period	Exclude Cancelled status group.
Data limitation	610 products have no category ('uncategorized').
Requirement	BR-02

BQ-03: Which customer regions and states generate the most revenue and orders, and which have the lowest average order value?

Business purpose	Identify strong and weak markets (BR-03).
Measures	Item sales revenue; Order count; Average order value (AOV)
Dimensions	dim_customer (region > state > city)
Fact table(s)	fact_orders
Filters / period	Exclude Cancelled status group.
Data limitation	Customer location is the most recent known address (SCD Type 1).
Requirement	BR-03

BQ-04: How concentrated is revenue among sellers (share of the top 10% of sellers), and in which states are the top sellers located?

Business purpose	Assess dependence on a small seller base (BR-04).
Measures	Item sales revenue; Units sold
Dimensions	dim_seller (region > state); dim_date
Fact table(s)	fact_order_items
Filters / period	Exclude Cancelled status group.
Data limitation	No seller cost data: revenue only, not seller profit.
Requirement	BR-04

BQ-05: What is the payment method mix by number of payments and by amount paid, and how does it change over time?

Business purpose	Understand customer payment preferences (BR-05).
Measures	Amount paid; Payment count
Dimensions	dim_payment_type; dim_date
Fact table(s)	fact_payments
Filters / period	Purchase date; all statuses.
Data limitation	No separate payment date in source; order purchase date is used.
Requirement	BR-05

BQ-06: How does the average credit-card payment value change with the number of instalments chosen?

Business purpose	Understand instalment behaviour for high-value purchases (BR-05).
Measures	Amount paid; Average instalments
Dimensions	dim_payment_type
Fact table(s)	fact_payments
Filters / period	payment_type_code = 'credit_card'.
Data limitation	Instalment interest is not identified in the data.
Requirement	BR-05

BQ-07: What share of orders end in each status, and how does the cancellation rate change by month?

Business purpose	Monitor operational failures (BR-06).
Measures	Order count; Cancellation rate
Dimensions	dim_order_status (group > status); dim_date
Fact table(s)	fact_orders
Filters / period	Purchase date.
Data limitation	Status is the final snapshot of the dataset; status history is not available.
Requirement	BR-06

BQ-08: What is the on-time delivery rate and the average delivery time by customer state and by month?

Business purpose	Find regions and periods with poor logistics performance (BR-07).
Measures	On-time delivery rate; Average delivery days; Average delivery delay
Dimensions	dim_customer (state); dim_date (purchase or delivery role)
Fact table(s)	fact_orders
Filters / period	is_delivered = 1.
Data limitation	8 delivered orders lack a delivery date and are excluded.
Requirement	BR-07

BQ-09: Which fulfilment stage (approval, seller handling, carrier transit) contributes most to total delivery time, and does this differ by customer region?

Business purpose	Target the right part of the process for improvement (BR-07).
Measures	days_purchase_to_approval; days_approval_to_carrier; days_carrier_to_customer
Dimensions	dim_date; dim_customer (region)
Fact table(s)	fact_orders
Filters / period	is_delivered = 1.
Data limitation	1,359 orders have a carrier date earlier than the approval date; these negative stage durations are flagged in Phase 2 data-quality checks and excluded from stage averages.
Requirement	BR-07

BQ-10: What is the distribution of review scores, and how does the average score and low-score share change by month?

Business purpose	Monitor customer satisfaction (BR-08).
Measures	Average review score; Low-score share
Dimensions	dim_date (review creation role)
Fact table(s)	fact_reviews
Filters / period	Review creation date.
Data limitation	Comment text is not analysed (Portuguese free text); only presence of a comment.
Requirement	BR-08

BQ-11: Do orders delivered late receive lower average review scores than orders delivered on time?

Business purpose	Quantify the association between delivery and satisfaction (BR-09).
Measures	avg_review_score; is_late; On-time delivery rate
Dimensions	dim_customer; dim_date
Fact table(s)	fact_orders
Filters / period	is_delivered = 1 AND review_count > 0.
Data limitation	Shows association, not causation.
Requirement	BR-09

BQ-12: What share of customers made more than one purchase, and how much revenue comes from repeat orders?

Business purpose	Measure customer retention (BR-10).
Measures	Repeat customer rate; Item sales revenue
Dimensions	dim_customer
Fact table(s)	fact_orders
Filters / period	Exclude Cancelled status group for revenue.
Data limitation	Only about 3% of customers are repeat buyers; no demographic data exists.
Requirement	BR-10

BQ-13: How large is freight relative to item price by product category, weight band and customer region?

Business purpose	Identify where shipping cost may discourage purchases (BR-11).
Measures	Freight-to-price ratio; Freight revenue
Dimensions	dim_product (category, weight band); dim_customer (region)
Fact table(s)	fact_order_items
Filters / period	Exclude Cancelled status group.
Data limitation	Freight charged to customer; actual carrier cost is not available.
Requirement	BR-11

4. Question groups

Area	Questions
Sales performance and trend	BQ-01, BQ-02
Geography and sellers	BQ-03, BQ-04
Payments	BQ-05, BQ-06
Orders and delivery	BQ-07, BQ-08, BQ-09
Satisfaction	BQ-10, BQ-11
Customers	BQ-12
Logistics cost	BQ-13

5. Questions that cannot be answered with this dataset

Analysis	Reason	Alternative used
Profit / margin by product or seller	No cost of goods, commission or carrier cost columns.	Revenue, freight-to-price ratio, AOV.
Inventory movement / stock snapshots	No stock level or warehouse data.	Units sold and demand trend by category.
Promotion / campaign effectiveness	No promotion, coupon campaign or marketing channel data.	Voucher usage share in payments (descriptive only).
Customer demographics (age, gender, income)	Not in source.	Behavioural segmentation: region, order frequency, spend.
Product-level or seller-level review analysis	Reviews are given per order, not per item; 1.3% of orders have several sellers.	Order-level review analysis; seller view limited to single-seller orders if required later.
Order status history	Only the final status is provided.	Milestone timestamps in fact_orders.

Results for these questions will be produced in Phase 3 from the loaded warehouse. This document does not contain any findings.